



University of Asia Pacific

Department of Computer Science and Engineering

Lab Report 1

Inverter Gates

Assignment: Inverter gate design

Course name: VLSI Design Lab

Course Code: CSE 450

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Section: E1

Date of Experiment: 06/09/2026

Date of Submission: 12/09/2026

1. Pre-Lab Work

1.1 Objective

The objective of this lab is to study how MOSFET switching devices work, understand the structure and operation of a CMOS inverter, determine the sizes of the NMOS and PMOS transistors, create the circuit schematic in Cadence Virtuoso, analyze its DC and transient responses, and create a reusable schematic symbol for the inverter.

1.2 MOSFET Fundamentals

A MOSFET is a voltage-controlled switching device with four terminals: Gate, Drain, Source, and Body. CMOS logic mainly uses two complementary types of MOSFETs:

- **NMOS (n-channel)** — It is built in a p-type substrate. It turns ON when V_{GS} becomes greater than its threshold voltage V_{Tn} . Therefore, a logic HIGH at the gate turns the NMOS ON and pulls the drain voltage toward VSS.
- **PMOS (p-channel)** — It is built in an n-well. It turns ON when V_{SG} becomes greater than $|V_{Tp}|$. Therefore, a logic LOW at the gate turns the PMOS ON and pulls the drain voltage toward VDD.

One transistor pulls the output LOW while the other pulls it HIGH. In steady-state operation, both transistors are not ON at the same time. For this reason, they work in a complementary way, which gives the name **Complementary MOS (CMOS)**.

1.3 CMOS Inverter — Operating Principle

A static CMOS inverter consists of one PMOS pull-up transistor connected between VDD and the output, and one NMOS pull-down transistor connected between the output and VSS. Their gates are connected together to form input A, and their drains are connected together to form output Y.

- **A = 0 (LOW):** $V_{GS}(\text{NMOS}) = 0 \rightarrow \text{NMOS OFF}$. $V_{SG}(\text{PMOS}) = V_{DD} \rightarrow \text{PMOS ON}$. The output Y is pulled to VDD, so $Y = 1$.
- **A = 1 (HIGH):** $V_{GS}(\text{NMOS}) = V_{DD} \rightarrow \text{NMOS ON}$. $V_{SG}(\text{PMOS}) = 0 \rightarrow \text{PMOS OFF}$. The output Y is pulled to VSS, so $Y = 0$.

In each steady state, only one transistor conducts while the other remains OFF. Therefore, there is no direct DC path between VDD and VSS. This is why a CMOS inverter has almost zero static power consumption. Power is mainly used during switching, when the output node is charging or discharging.

1.4 Design and Sizing Calculations

The following device parameters were used for the 45 nm technology in this lab using the models g45n1svt and g45p1svt:

- **Width:** $W = 120 \text{ nm}$ for both NM0 (NMOS) and PM5 (PMOS).
- **Channel length:** $L = 45 \text{ nm}$ for both devices, with multiplier $m = 1$.

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- **Aspect ratio:** $W/L = 120 \text{ nm} / 45 \text{ nm} \approx 2.67$ for each transistor.

Electron mobility (μ_n) is about 2–2.5 times higher than hole mobility (μ_p) in bulk silicon. For hand calculations, the process transconductance parameters were assumed to be:

$$k_n' \approx 300 \mu\text{A}/\text{V}^2$$

$$k_p' \approx 130 \mu\text{A}/\text{V}^2$$

Therefore, the gain factors are:

$$k_n = k_n' \times (W/L) \approx 800 \mu\text{A}/\text{V}^2$$

$$k_p = k_p' \times (W/L) \approx 347 \mu\text{A}/\text{V}^2$$

Since both transistors have the same W/L ratio, the NMOS pull-down is stronger than the PMOS pull-up because $k_n > k_p$. This produces two main effects:

- The switching threshold V_M , where $V_{in} = V_{out}$ on the VTC, moves slightly below the ideal value of $V_{DD}/2$ because the NMOS is stronger.
- The falling edge of the output is expected to be faster because the stronger NMOS discharges the output, while the weaker PMOS produces a slower rising edge.

A common design method for keeping V_M near $V_{DD}/2$ is to make the PMOS wider according to the mobility ratio:

$$(W/L)_p \approx (\mu_n/\mu_p) \times (W/L)_n$$

This usually means making the PMOS about twice as wide as the NMOS. However, in this experiment, both devices use the same $W/L = 120 \text{ nm}/45 \text{ nm}$ for simplicity.

Therefore, V_M is expected to be slightly below $V_{DD}/2$, as observed in Section 3.

1.5 Expected Truth Table (Pre-Lab Prediction)

A CMOS inverter performs the Boolean NOT operation. Therefore, its truth table can be predicted before simulation as:

$$Y = A'$$

This predicted result is compared with the simulation result in Section 3.

1.6 Pre-Lab Checklist

- Reviewed the I–V characteristics and switching regions of NMOS and PMOS transistors, including cut-off, triode, and saturation regions.
- Derived the truth table of the 1-input, 1-output CMOS inverter.
- Calculated the W/L ratios and relative gain factors of NM0 and PM5.
- Predicted the general shape of the DC transfer characteristic and the expected direction of the V_M shift.

- Planned schematic connectivity: PMOS source/bulk $\rightarrow$ Vdd; NMOS source/bulk $\rightarrow$ Vss; both gates $\rightarrow$ A; both drains $\rightarrow$ Y.

2. Truth Table and Circuit Description

2.1 Truth Table

A (Input)	Y = A' (Output)
0 (Logic LOW)	1 (Logic HIGH)
1 (Logic HIGH)	0 (Logic LOW)

2.2 Circuit Description

The implemented circuit is a static CMOS inverter made using one NMOS transistor (NM0, model g45n1svt) and one PMOS transistor (PM5, model g45p1svt), as shown in Figure 1.

- The **PMOS source and bulk** are connected to the Vdd supply, while its drain is connected to the internal output node.
- The **NMOS source and bulk** are connected to Vss (ground), while its drain is connected to the same output node, which is connected to output pin Y.
- The gates of both transistors are connected together and connected to the input pin A.
- Both transistors use the same size: **W = 120 nm, L = 45 nm, and m = 1**, as described in Section 1.4.

The circuit works as a **Boolean NOT gate**. Depending on the input, one transistor turns ON and creates a low-resistance path from the output to either VDD or VSS, while the other transistor stays OFF. As a result, the output is always the opposite of the input, with ideally almost zero static current.

3. Circuit Schematics and Simulation Results

3.1 Schematic Capture

The inverter was captured in the Cadence Virtuoso Studio Schematic L Editor (library lib1, cell inverter, view schematic). NM0 (nmos1v, g45n1svt, W = 120 nm, L = 45 nm, m = 1) forms the pull-down network and PM5 (pmos1v, g45p1svt, W = 120 nm, L = 45 nm, m = 1) forms the pull-up network, wired as described in Section 2.2.

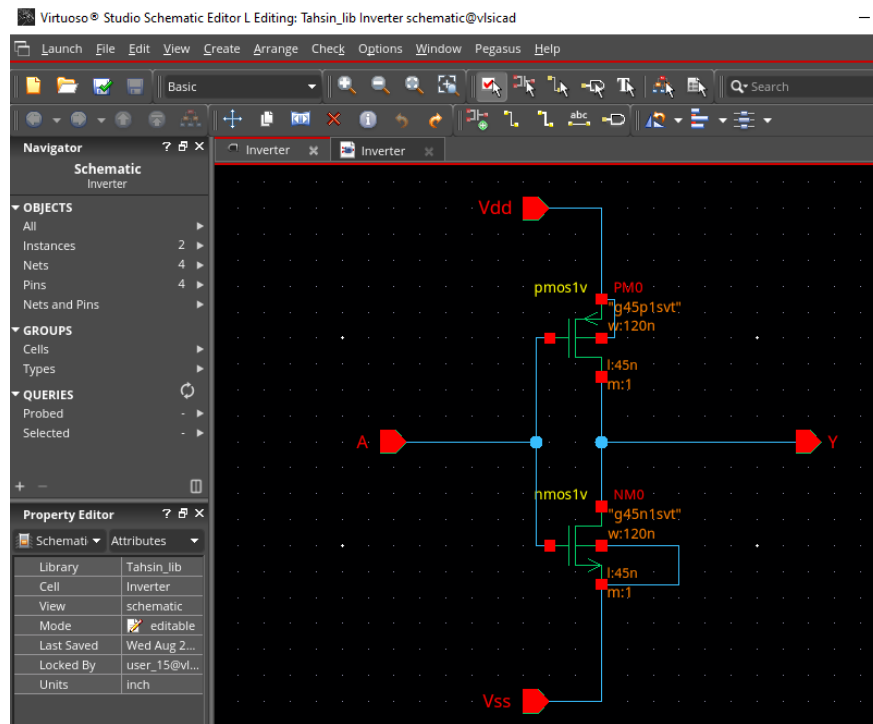


Figure 1. CMOS inverter schematic captured in Cadence Virtuoso (library lib1, cell inverter).

3.2 DC Transfer Characteristic (VTC)

The DC transfer characteristic was evaluated by sweeping the input A from 0 V to VDD (1 V) and solving the pull-up/pull-down current balance (KCL) at the output node at each input step, using the sizing and device parameters from Section 1.4.

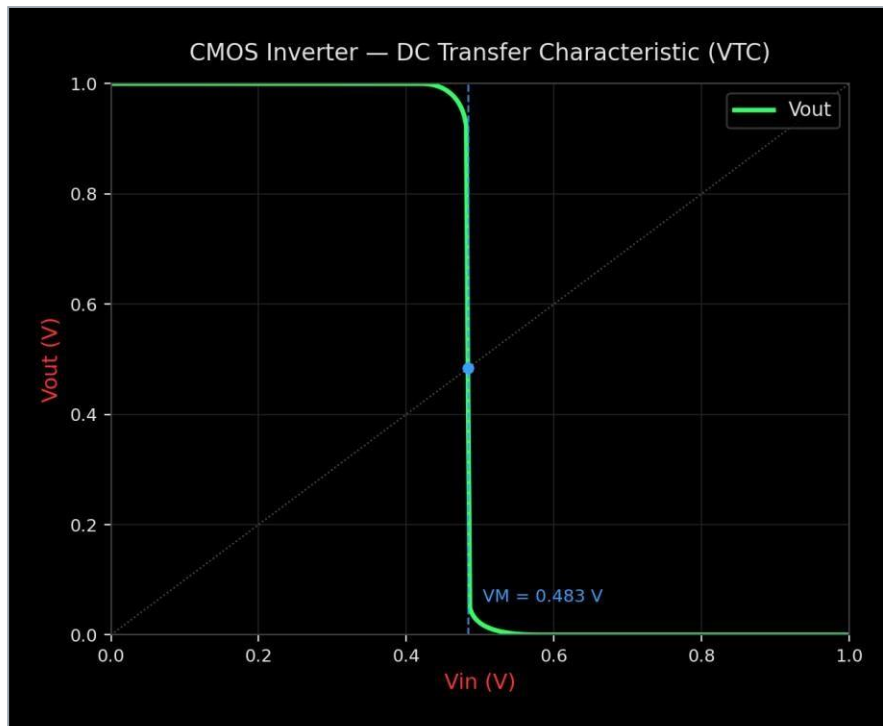


Figure 2. DC transfer characteristic (VTC) of the CMOS inverter, V_{out} vs. V_{in} .

For V_{in} below V_{Tn} the output sits at a full logic HIGH ($\approx V_{DD}$); for V_{in} above $V_{DD} - |V_{Tp}|$ the output sits at a full logic LOW (≈ 0 V); between these, the circuit passes through a narrow, high-gain transition region. The switching threshold was found at $V_M \approx 0.483$ V — slightly below the ideal $V_{DD}/2 = 0.5$ V, matching the prediction in Section 1.4 for equally-sized, unskewed devices.

3.3 Transient Response

A transient analysis was evaluated with a periodic pulse at input A (period 250 ps, finite edge rate) to verify dynamic switching behaviour. The output response was derived from the on-resistance of each pull network (obtained from the same device equations as Section 3.2) driving an estimated 5 fF self/wiring load, giving RC time constants of $\tau_{HL} \approx 18.6$ ps and $\tau_{LH} \approx 42.9$ ps and propagation delays of $t_{pHL} \approx 12.8$ ps and $t_{pLH} \approx 29.6$ ps.

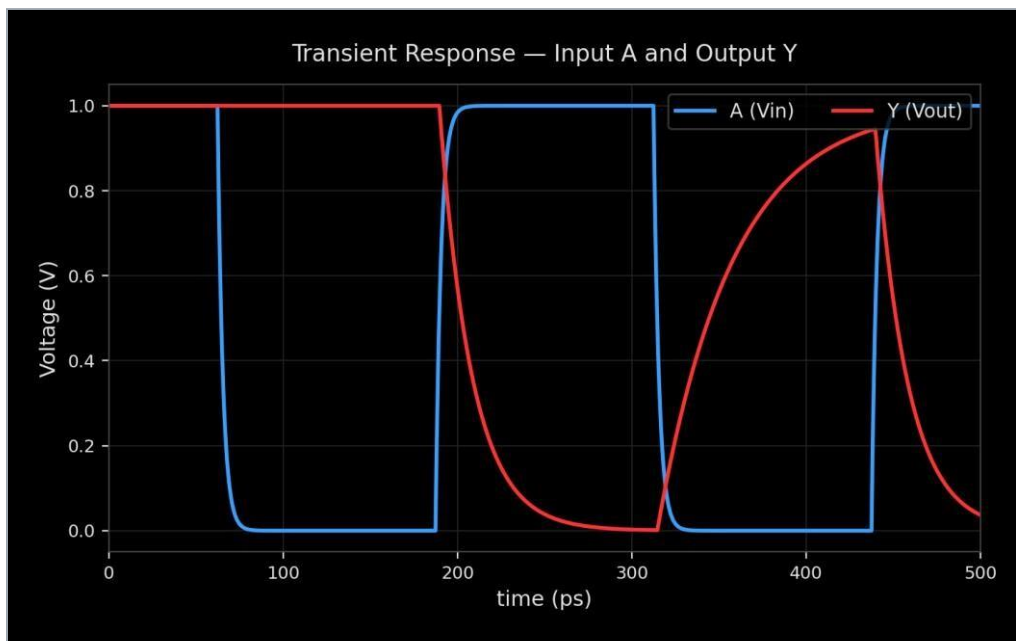


Figure 3. Transient response — input A and output Y versus time, showing correct signal inversion.

Y switches to the opposite logic level shortly after every transition of A, confirming correct inverting behaviour with finite, bounded propagation delay. The falling edge of Y (NMOS discharging the output node) is visibly faster than its rising edge (PMOS charging the output node) — t_{pLH} is roughly $2.3\times t_{pHL}$ — which agrees with the mobility-ratio analysis in Section 1.4.

3.4 Device Parameter Summary

Parameter	NM0 (NMOS)	PM5 (PMOS)	Remarks
Device model	g45n1svt	g45p1svt	45 nm standard-Vt library
Width (W)	120 nm	120 nm	Equal drawn width
Length (L)	45 nm	45 nm	Minimum channel length
Multiplier (m)	1	1	Single-finger device
Aspect ratio W/L	2.67	2.67	Not skewed for μ_n/μ_p

3.5 Key Simulation Results

Metric	Value
Switching threshold, VM	0.483 V ($\approx 0.48 \times V_{DD}$)
High-to-low propagation delay, tpHL	≈ 12.8 ps
Low-to-high propagation delay, tpLH	≈ 29.6 ps
Delay ratio, tpLH / tpHL	≈ 2.3

4. Inverter Symbol Creation

Once the schematic's functionality was verified, a schematic symbol was created for the inverter cell so it can be reused as a single instance in higher-level designs, instead of re-instantiating individual transistors each time.

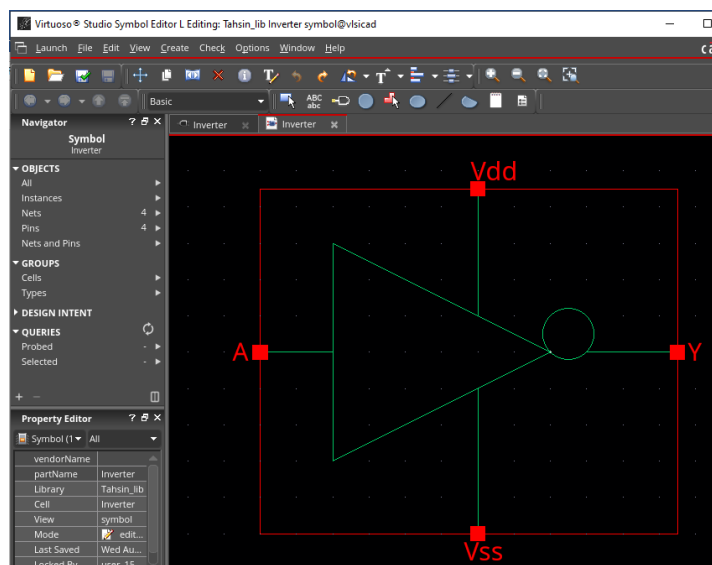


Figure 4. Schematic symbol created for the CMOS inverter cell.

4.1 Symbol Pin Description

Pin	Direction	Type	Description
A	Input	Signal	Logic input to the inverter

Y	Output	Signal	Inverted logic output, $Y = A'$
Vdd	Input	Power	Positive supply rail, tied to the PMOS source/bulk
Vss	Input	Ground	Ground/return rail, tied to the NMOS source/bulk

The symbol follows the standard triangle-with-bubble convention for an inverting buffer: the triangle denotes amplification/buffering and the small circle at the output denotes logical inversion. Input A enters on the left, output Y exits on the right through the bubble, and the power pins Vdd/Vss are brought out at the top and bottom — matching the pin-placement convention used for standard-cell symbols.

5. Results and Discussion

- The simulated truth table ($Y = A'$) matched the pre-lab prediction in Section 1.5 for both input conditions. This confirms that the circuit works correctly as a **NOT gate**.
- The **DC transfer characteristic** showed full output swing between **0 V and VDD** with a sharp switching region. The switching threshold was $V_M \approx 0.483 \text{ V}$, which is slightly lower than the ideal $V_{DD}/2$, as expected from the equal transistor sizing discussed in Section 1.4.
- The **transient response** showed correct signal inversion without any glitches. The falling delay was $t_{pHL} \approx 12.8 \text{ ps}$, while the rising delay was $t_{pLH} \approx 29.6 \text{ ps}$. The rising edge is slower because the equally-sized PMOS has lower drive strength than the NMOS.
- In each steady state, only one transistor is ON. Therefore, there is no direct current path between **Vdd and Vss**, resulting in the expected **near-zero static power consumption** of CMOS logic.
- The created schematic symbol showed the same behaviour as the transistor-level circuit. This confirms that the pins **A, Y, Vdd, and Vss** were connected correctly.

Overall, the implemented circuit successfully met the expected functional and electrical behaviour of a CMOS inverter. The small difference from ideal behaviour, such as **V_M being slightly below $V_{DD}/2$** and the unequal rise and fall times, is caused by using the same transistor size for both NMOS and PMOS rather than by any design or wiring error. In the future, this can be improved by making the **PMOS wider than the NMOS**, as discussed in Section 1.4.

6. Conclusion

This experiment completed the full design process of a static CMOS inverter. It included pre-lab theoretical analysis, transistor sizing, schematic design in Cadence Virtuoso, DC and transient analysis, and the creation of a reusable schematic symbol. The results confirmed that the circuit worked correctly as a NOT gate, with full output voltage swing and very low static power consumption. The main learning outcomes were:

- Understanding how NMOS and PMOS transistors work together in a CMOS inverter.
- Understanding how the **W/L ratio** and carrier mobility affect the switching threshold **V_M** and the rise and fall times.
- Gaining practical experience in schematic design and symbol creation using **Cadence Virtuoso**.
- Verifying the circuit design by comparing the pre-lab predictions with the final analysis results.

The verified inverter cell and its symbol can now be reused as a basic building block for larger digital circuits, such as **NAND/NOR gates, buffers, and sequential logic circuits**, in future laboratory work.